Comparison of Parametric and Non-Parametric Methods for Predicting BMI

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Introduction

Obesity is a disease of excessive body fat, a major concern for overall health status. Body Mass Index (BMI) is often used as a proxy to measure obesity levels. By assessing body weight relative to height, the continuous measure indicates whether an individual is overweight and falls into the obesity range. Overweight individuals are associated with higher physical and mental health risks, such as diabetes, cardiovascular disease, high blood pressure, and anxiety. This is driven by a combination of factors like genetics, environment, and poor diet choices. The relationship between such demographics, lifestyle factors, and BMI is complex. Simple regression models may not fully capture the complexity of BMI prediction. This study aims to explore whether increasing model flexibility affects model performance for BMI using an obesity-related dataset containing demographic and behavioral variables. The analysis compares parametric and nonparametric regression approaches. Specifically, multiple linear regression, polynomial regression, generalized additive models (GAMs), and random forest regression are evaluated and compared for prediction accuracy.

Material and Methods

The dataset on obesity levels was obtained from the UC Irvine Machine Learning Repository. It includes demographics, eating habits, and physical condition of 2111 participants from Mexico, Peru, and Colombia.

Explanatory Variables:

- Gender
- Age
- family_history_with_overweight
- FAVC (Frequent consumption of high-calorie foods)
- FCVC (Frequency of vegetable consumption)
- NCP (Number of main meals consumed daily)
- CAEC (Frequency of eating food between meals)
- SMOKE (Smoking status)
- CH2O (Daily water consumption)
- SCC (Whether the participant monitors daily caloric intake)
- FAF (Frequency of physical activity)

- TUE (Time spent using technological devices)
- CALC (Frequency of alcohol consumption)
- MTRANS (Usual mode of transportation)

Categorical variables were converted into factors for analysis.

Response Variable

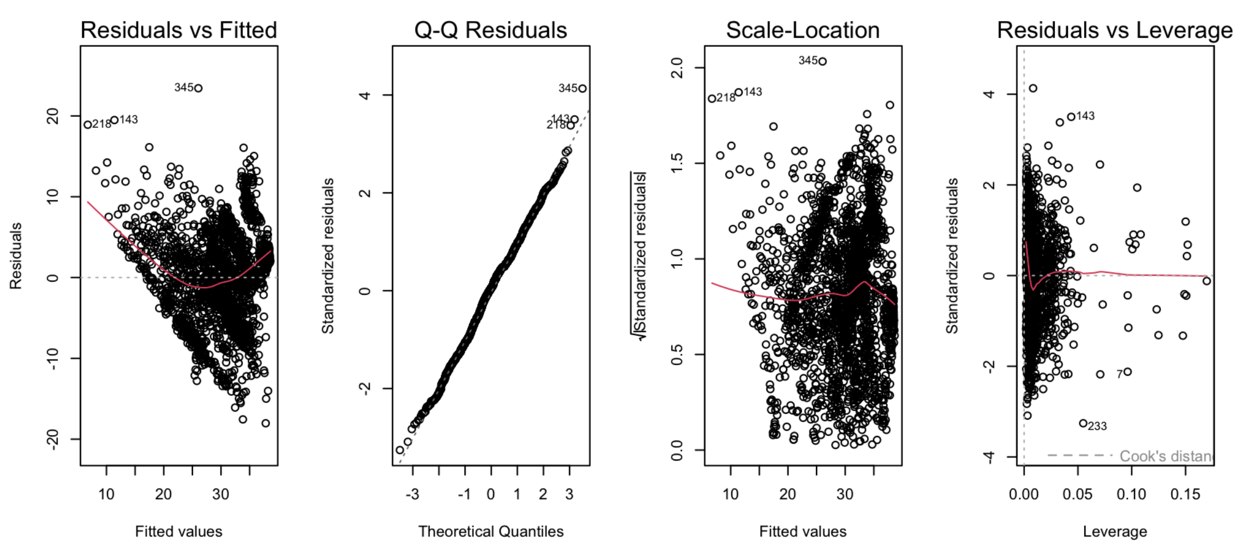
BMI was calculated using weight and height variables from the dataset.

$$BMI = Weight/Height^2$$

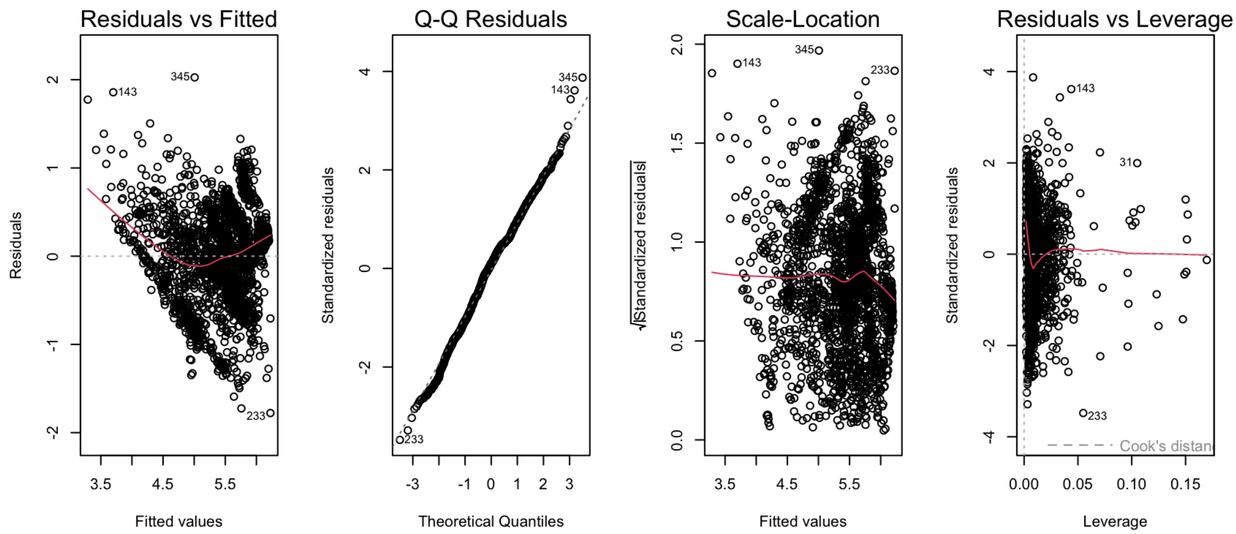
Statistical Analysis

Summary statistics were used to analyze the relationships between BMI and the demographic variables. Histograms and diagnostic plots were used to assess assumptions. BMI distribution was right-skewed and obtained a λ of 0.5 from the Box-Cox transformation. However, the transformation did not significantly improve model assumptions and was not transformed for interpretability for model comparison. (See diagnostic plots below)

Linear Model of Original BMI



Linear Model of Square Root Transformed BMI



The data was randomly divided into training and validation sets, using an 80-20 split to evaluate the prediction accuracy across models. Training R^2 values and test root mean square error (RMSE) were used to compare model performance between regression approaches.

Multiple linear regression was used as the baseline parametric model. All variables were fitted into the model. This model assumes linear relationships between predictors and BMI and served as a reference point for evaluating more flexible methods. Assumptions and multicollinearity were examined.

Exploratory plots (See Appendix) and residual diagnostics from linear regression suggested nonlinear relationships with BMI. To address this, polynomial regression models with degree = 2 were fit for selected variables: Age, NCP, and TUE

Generalized additive models (GAMs) were used to allow more flexible nonlinear relationships between predictors and BMI. Splines were fit for selected variables: Age, NCP, and TUE. gam from mgcv was used for automatic selection.

Random Forest regression was included as a flexible nonparametric regression method. Since Random Forest is less constrained by assumptions such as linearity and is able to capture complex relationships, it is used to compare model predictive accuracy.

Results

Multiple Linear Regression

Model:

BMI ~ GenderMale + Age + family_history_with_overweightYes + FAVCYES + FCVC + NCP + CAECFrequently + CAECno + CAECSometimes + SMOKEYes + CH2O + SCCYes + FAF + TUE + CALCFrequently + CALCno + CALCSometimes + MTRANSBike + MTRANSMotorbike + MTRANSPublic_Transportation + MTRANSWalking

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Residuals:
     Min       1Q   Median       3Q      Max 
-19.0977  -3.8964   0.2581   3.5443  24.1501 

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)                        3.18289    6.00039   0.530 0.595872    
GenderMale                        -0.58249    0.30280  -1.924 0.054568 .  
Age                                0.33485    0.03033  11.042  < 2e-16 ***
family_history_with_overweightyes  6.60197    0.40537  16.286  < 2e-16 ***
FAVCyes                            1.55623    0.46279   3.363 0.000789 ***
FCVC                               3.43758    0.27717  12.402  < 2e-16 ***
NCP                                0.47543    0.18288   2.600 0.009412 ** 
CAECFrequently                    -3.98559    0.92649  -4.302 1.79e-05 ***
CAECno                             2.58316    1.23883   2.085 0.037207 *  
CAECSometimes                      3.37697    0.85017   3.972 7.43e-05 ***
SMOKEyes                          -0.60419    0.98671  -0.612 0.540405    
CH2O                               0.73239    0.24172   3.030 0.002484 ** 
SCCyes                            -2.23088    0.70742  -3.154 0.001642 ** 
FAF                               -0.82568    0.17459  -4.729 2.44e-06 ***
TUE                               -0.52282    0.24640  -2.122 0.033999 *  
CALCFrequently                    -4.92842    5.80724  -0.849 0.396188    
CALCno                            -5.59558    5.76215  -0.971 0.331643    
CALCSometimes                     -3.28967    5.76737  -0.570 0.568488    
MTRANSBike                         2.19200    2.35537   0.931 0.352175    
MTRANSMotorbike                    2.81584    2.05009   1.374 0.169776    
MTRANSPublic_Transportation        4.78197    0.43961  10.878  < 2e-16 ***
MTRANSWalking                      1.74240    0.94107   1.852 0.064273 .  
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Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.661 on 1666 degrees of freedom
Multiple R-squared:  0.5084,	Adjusted R-squared:  0.5022 
F-statistic: 82.05 on 21 and 1666 DF,  p-value: < 2.2e-16
```

The multiple linear regression of the full prediction model fitted on the training data explained 50.84% of the variation in BMI. ($R^2 = 0.5084$) with a test RMSE of 5.862164. Variance inflation factors were relatively small. No multicolinearity was detected. An outlier is detected.

Most predictors were statistically significant, except for smoking status and frequency of alcohol consumption, and categories of using bike and motorbike as their mode of transport.

Factors such as being male, those who eat between meals, participate in tracking calories, frequently exercise, and spend more time on technological devices show a significant association with having a lower BMI. In contrast, factors such as having a family history of overweight, frequent consumption of high-calorie food, an increase in the number of meals per day, high consumption of water, and use of public transportation are associated with higher BMI.

Polynomial Regression

Model:

BMI ~ GenderMale + poly(Age,2) + family_history_with_overweightYes + FAVCYES + FCVC + poly(NCP,2) + CAECFrequently + CAECno + CAECSometimes + SMOKEYes + CH2O + SCCYes + FAF + poly(TUE,2) + CALCFrequently + CALCno + CALCSometimes + MTRANSBike + MTRANSMotorbike + MTRANSPublic_Transportation + MTRANSWalking

```
Residuals:
     Min       1Q   Median       3Q      Max 
-16.5608  -3.7527   0.2897   3.4108  24.0870 

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)                        16.9151     5.7137   2.960 0.003115 ** 
GenderMale                         -0.7904     0.2939  -2.690 0.007223 ** 
family_history_with_overweightyes   5.8697     0.3974  14.770  < 2e-16 ***
FAVCyes                             1.7242     0.4458   3.867 0.000114 ***
FCVC                                2.9868     0.2709  11.025  < 2e-16 ***
CAECFrequently                     -4.6963     0.8932  -5.258 1.65e-07 ***
CAECno                              1.8525     1.1940   1.551 0.120978    
CAECSometimes                       1.9708     0.8277   2.381 0.017375 *  
SMOKEyes                            0.4305     0.9546   0.451 0.652085    
CH2O                                0.7772     0.2329   3.337 0.000865 ***
SCCyes                             -1.6410     0.6891  -2.381 0.017359 *  
FAF                                -0.6374     0.1695  -3.760 0.000176 ***
CALCFrequently                     -6.8093     5.5909  -1.218 0.223428    
CALCno                             -6.7388     5.5450  -1.215 0.224428    
CALCSometimes                      -5.1622     5.5513  -0.930 0.352546    
MTRANSBike                          2.3477     2.2710   1.034 0.301399    
MTRANSMotorbike                     2.7996     1.9738   1.418 0.156273    
MTRANSPublic_Transportation         4.1616     0.4286   9.709  < 2e-16 ***
MTRANSWalking                       1.9873     0.9098   2.184 0.029079 *  
poly(Age, 2)1                      82.1085     7.6787  10.693  < 2e-16 ***
poly(Age, 2)2                     -42.4292     5.8259  -7.283 5.02e-13 ***
poly(NCP, 2)1                      13.6989     5.7564   2.380 0.017435 *  
poly(NCP, 2)2                     -39.7017     5.9219  -6.704 2.77e-11 ***
poly(TUE, 2)1                     -11.8671     5.9569  -1.992 0.046517 *  
poly(TUE, 2)2                     -31.8749     5.8900  -5.412 7.15e-08 ***
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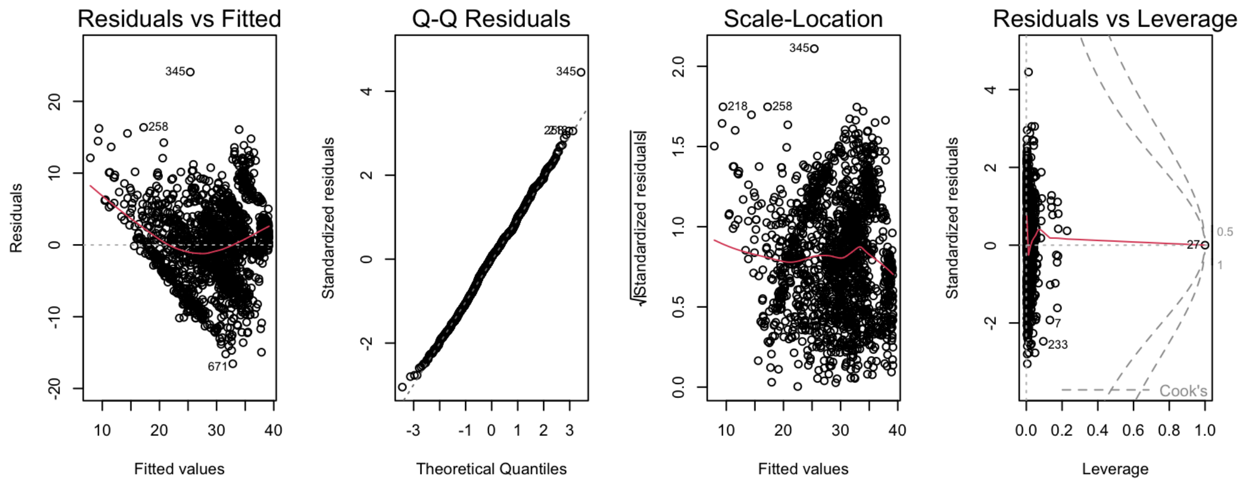
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.445 on 1663 degrees of freedom
Multiple R-squared:  0.5461,	Adjusted R-squared:  0.5395 
F-statistic: 83.35 on 24 and 1663 DF,  p-value: < 2.2e-16
```



To capture the nonlinearity of the data, a polynomial regression was fitted. The model explained 54.61% of the variation in DDS. ($R^2 = 0.5461$) with a test RMSE of 5.61621. General trends of significant predictors were consistent with multiple linear regression. With the addition of polynomial terms (degree = 2), a significant nonlinear relationship between BMI and the variables age, number of meals, and time spent on technological devices was detected. They first increase, then decrease BMI. However, diagnostic plots show that adding a polynomial term was not able to fully capture the nonlinearity in the fitted residuals.

Generalized Additive Model

Model:

BMI ~ GenderMale + s(Age) + family_history_with_overweightYes + FAVCYES + FCVC + s(NCP) + CAECFrequently + CAECno + CAECSometimes + SMOKEYes + CH2O + SCCYes + FAF + s(TUE) + CALCFrequently + CALCno + CALCSometimes + MTRANSBike + MTRANSMotorbike + MTRANSPublic_Transportation + MTRANSWalking

```
Parametric coefficients:
                                 Estimate Std. Error t value
(Intercept)                       17.7850     5.4033   3.291
GenderMale                        -0.4976     0.2869  -1.735
family_history_with_overweightyes  5.4426     0.3800  14.324
FAVCyes                            1.5171     0.4264   3.558
FCVC                               2.5701     0.2597   9.897
CAECFrequently                    -4.0886     0.8510  -4.804
CAECno                             2.7271     1.1439   2.384
CAECSometimes                      2.5204     0.7934   3.177
SMOKEyes                           0.2957     0.9136   0.324
CH2O                               0.4257     0.2285   1.863
SCCyes                            -1.4642     0.6588  -2.223
FAF                               -0.3903     0.1667  -2.341
CALCFrequently                    -6.1056     5.2835  -1.156
CALCno                            -5.1806     5.2435  -0.988
```

```
CALCSometimes                    -4.2142     5.2460  -0.803
MTRANSBike                        0.6749     2.1528   0.314
MTRANSMotorbike                   1.8587     1.8763   0.991
MTRANSPublic_Transportation       3.2922     0.4526   7.274
MTRANSWalking                     0.8291     0.8846   0.937
                                 Pr(>|t|)
(Intercept)                      0.001018 **
GenderMale                       0.083000 .
family_history_with_overweightyes  < 2e-16 ***
FAVCyes                          0.000385 ***
FCVC                              < 2e-16 ***
CAECFrequently                   1.69e-06 ***
CAECno                           0.017235 *
CAECSometimes                    0.001518 **
SMOKEyes                         0.746188
CH2O                             0.062601 .
SCCyes                           0.026377 *
FAF                              0.019334 *
CALCFrequently                   0.248015
CALCno                           0.323292
CALCSometimes                    0.421906
MTRANSBike                       0.753930
MTRANSMotorbike                  0.322015
MTRANSPublic_Transportation      5.38e-13 ***
MTRANSWalking                    0.348763
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Approximate significance of smooth terms:
         edf Ref.df     F p-value
s(Age) 7.772  8.627 22.21  <2e-16 ***
s(NCP) 8.305  8.863 17.13  <2e-16 ***
s(TUE) 8.303  8.860 13.39  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

R-sq.(adj) =  0.591   Deviance explained = 60.1%
GCV = 27.026  Scale est. = 26.332    n = 1688
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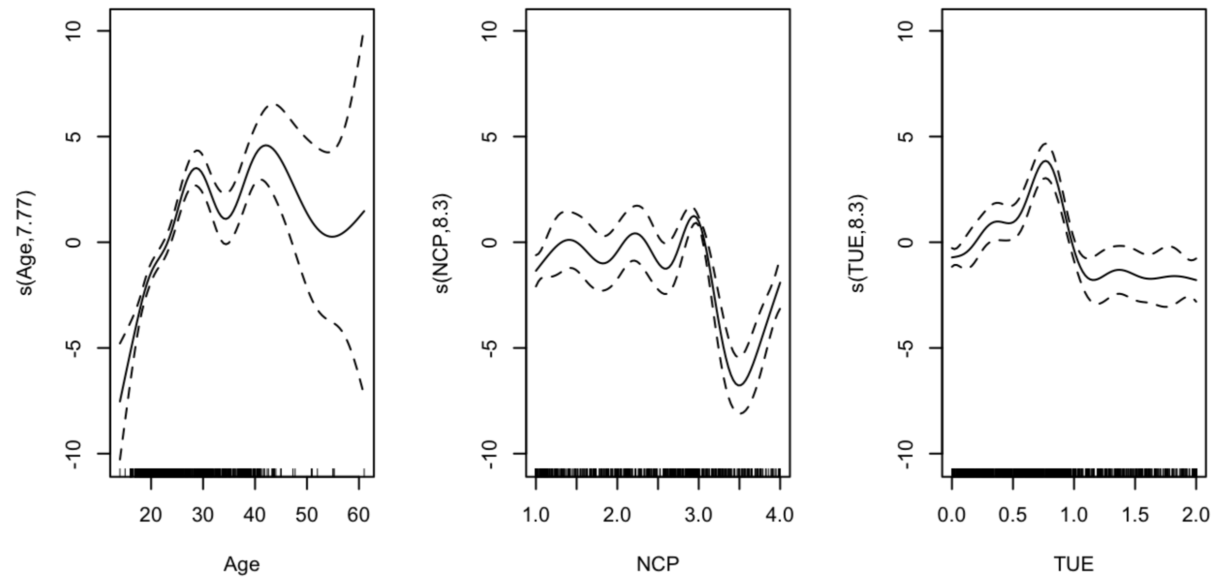


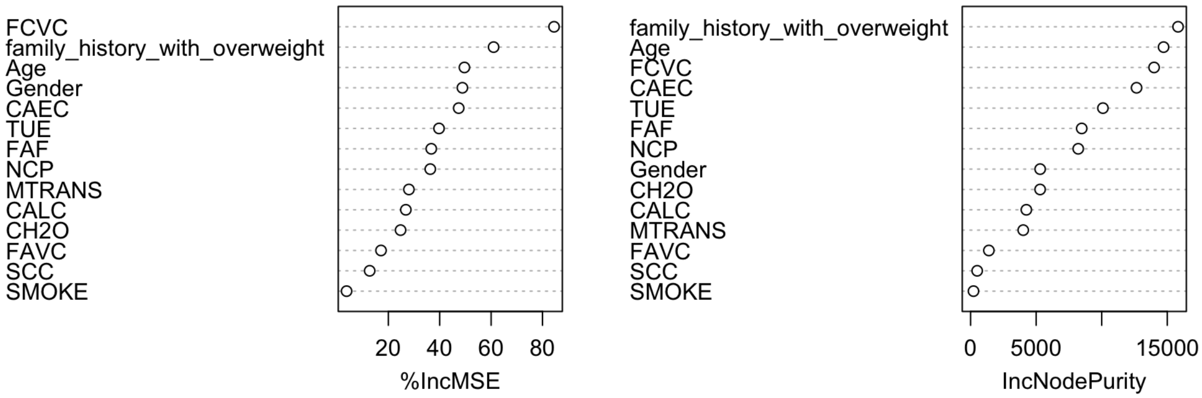
The GAM with spline was fitted on the training data explained 59.1% of the variation in BMI with a test RMSE of 5.147785. Significant predictors were consistent with linear and polynomial regression, with the addition of significance for splines for Age and number of meals, and time spent on technological devices. This suggests that there may more complex nonlinear relationship than could be captured by polynomial regression.

Random Forest

```
Call:
 randomForest(formula = BMI ~ ., data = train, importance = T)
              Type of random forest: regression
                    Number of trees: 500
No. of variables tried at each split: 4

          Mean of squared residuals: 7.856692
                    % Var explained: 87.79

                               %IncMSE IncNodePurity
Gender                       48.830659     5304.4288
Age                          49.617392    14716.9767
family_history_with_overweight 60.977118    15821.0087
FAVC                         17.154380     1395.2664
FCVC                         84.502134    13992.5458
NCP                          36.332748     8208.9850
CAEC                         47.389082    12655.7464
SMOKE                         3.754619      222.4726
CH2O                         24.768133     5301.7673
SCC                          12.743848      503.7692
FAF                          36.714426     8475.3641
TUE                          39.728503    10091.4163
CALC                         26.817334     4253.2037
MTRANS                       28.016467     4014.3309
```



Random Forest fitted on the training data explained 87.79% of the variation in BMI with a test RMSE of 2.728341. Smoking and caloric intake were shown to have little importance, while family history of overweight and FCVC were shown to be the most important.

Method	R^2	RMSE
Multiple Linear Regression	0.5084	5.862164
Polynomial Regression	0.5461	5.61621
GAM with splines	0.591	5.147785
Random Forest	0.8779	2.728341

Overall, predictive performance increased as model flexibility increased. Predictive accuracy was dramatically improved for Random Forest, which outperformed all other methods. Non-parametric methods performed better than parametric methods.

Discussion

This project compared parametric, semiparametric, and nonparametric regression methods for BMI prediction. Exploratory plots and model diagnostics consistently suggested that several predictors exhibited nonlinear relationships with BMI. Nonparametric methods are shown to improve the prediction of BMI in this dataset.

The multiple linear regression model served as an important basis for interpretation but displayed residual curvature and mild assumption violations. Polynomial regression improved the fit by allowing quadratic relationships. Generalized additive models provided improvements by smoothing effects. Random forest regression achieved strong predictive performance by flexibly capturing nonlinear relationships and interactions. However, the increased flexibility faces a trade-off of reduced interpretability.

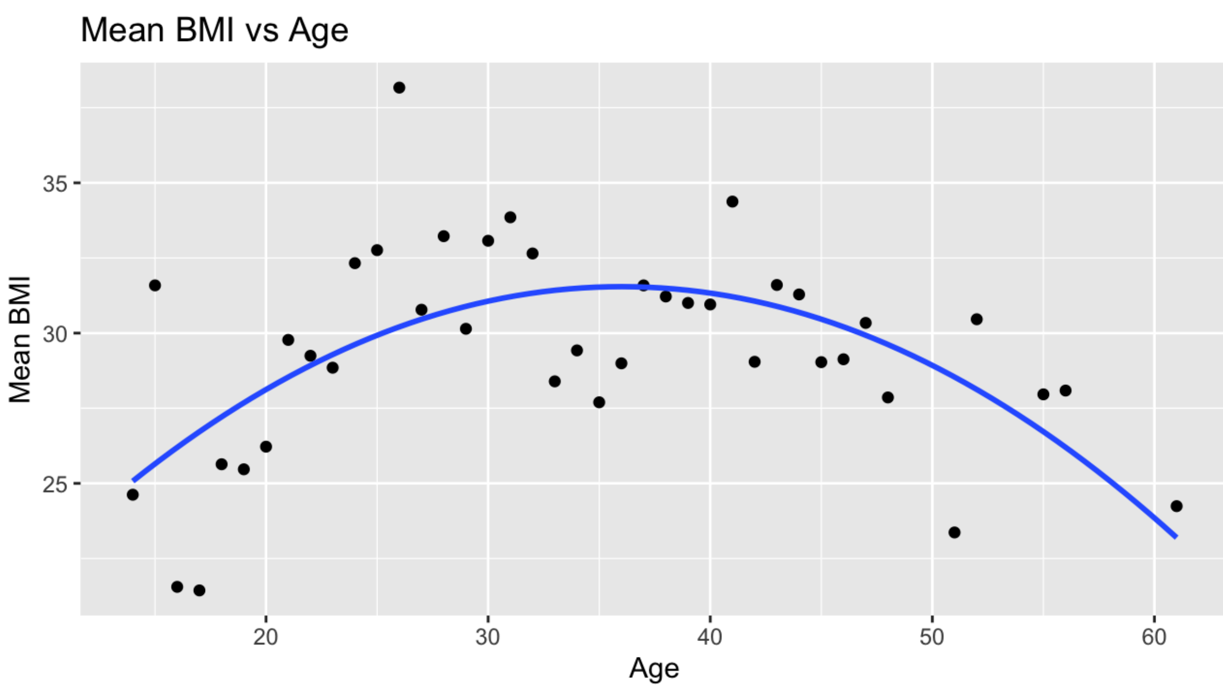
Several limitations should be acknowledged. The dataset is limited and may not generalize to broader populations. Future work includes exploring additional regression approaches, such as boosting. Additional predictors and external validation datasets may also improve predictive performance.

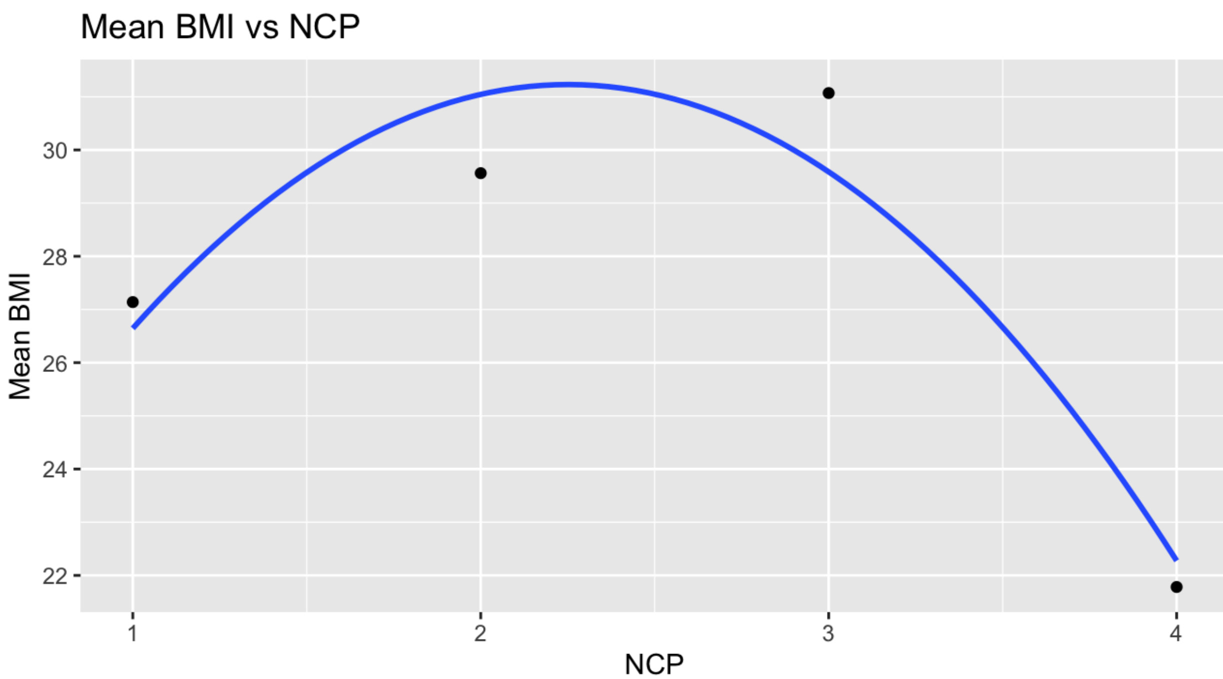
References

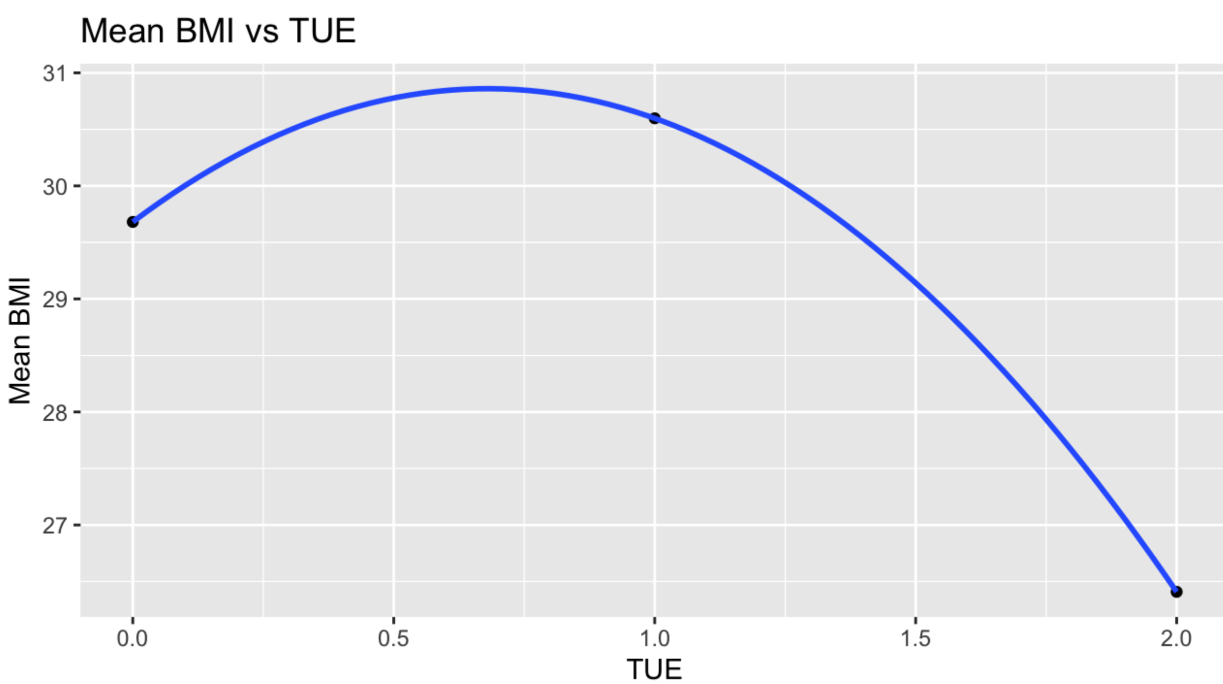
Estimation of Obesity Levels Based On Eating Habits and Physical Condition [Dataset]. (2019). UCI Machine Learning Repository. https://doi.org/10.24432/C5H31Z.

Appendix

Additional Graphs




Mean BMI vs NCP
Mean BMI
30
28
26
24
22
1
2
3
4
NCP


Mean BMI vs TUE
Mean BMI
31
30
29
28
27
0.0
0.5
1.0
1.5
2.0
TUE